

UNIT - 1

Introduction To MySQL

SQL (Structured Query Language) is used to manage and work with data in databases. It helps to create, store, retrieve, and modify data using database systems such as MySQL, PostgreSQL, and MsSQL.

Data Types

Category	Examples
Number	INT, BIGINT, FLOAT, DECIMAL
Date/Time	DATE, DATETIME, TIMESTAMP
Text	CHAR, VARCHAR, TEXT
Special	BLOB, BOOLEAN, JSON

Types of SQL Commands

DDL (Data Definition Language)

DDL is used to create and modify the structure of database objects.

- CREATE – Creates a table.
- ALTER – Modifies a table.
- DROP – Deletes a table.
- TRUNCATE – Removes all records from a table.

DML (Data Manipulation Language)

DML is used to add, modify, delete and retrieve data from tables.

- INSERT – Adds new data.
- UPDATE – Modifies existing data.
- DELETE – Removes data.
- SELECT – Retrieves or displays data.

TCL((Transaction Control Language)

TCL is used to manage transactions and control changes.

- COMMIT – Permanently saves changes.
- ROLLBACK – Undoes changes.
- SAVEPOINT – Creates a point for rollback.

Column Attributes

Attribute	Use
NOT NULL	Prevents NULL values
AUTO_INCREMENT	Generates numbers automatically
DEFAULT	Sets a default value
PRIMARY KEY	Uniquely identifies a record

Analytical Reporting Case Study:

#Create a database and use the database

```
create database movies_db;
use movies_db;

create table movies2
(
  rel_year int,
  mov_name varchar(100),
  act_name varchar(50),
  lang varchar(30),
  dir_name varchar(100)
);
```

#Insert the data's

```
INSERT INTO MOVIES2 VALUES (1960,'Kalathur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES2 VALUES (1962,'Kannum Karalum','Kamal Haasan','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES2 VALUES (1962,'Paadha Kaanikkai','Kamal Haasan','Tamil','K. Shankar');
INSERT INTO MOVIES2 VALUES (1962,'Paarthaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES2 VALUES (1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy');
INSERT INTO MOVIES2 VALUES (1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan');
INSERT INTO MOVIES2 VALUES (1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam');
INSERT INTO MOVIES2 VALUES (1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan');
INSERT INTO MOVIES2 VALUES (1971,'Anubhavangal Paalichakai','Mammootty','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES2 VALUES (1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan');
INSERT INTO MOVIES2 VALUES (1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES2 VALUES (1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan');
```

1.Retrieve all details from the movie table.

```
SELECT * FROM MOVIES2;
```

Solution:

	rel_year	mov_name	act_name	lang	dir_name
►	1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
	1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
	1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
	1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
	1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
	1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
	1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
	1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan
	1971	Anubhavgal Paalichakal	Mammootty	Malayalam	K. S. Sethumadhavan

2.Display only movie names and actor names.

```
SELECT mov_name, act_name FROM MOVIES2;
```

Solution:

	rel_year	mov_name	act_name	lang	dir_name
►	1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
	1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
	1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
	1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
	1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
	1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
	1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
	1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan

3.Display distinct movie languages.

```
SELECT DISTINCT lang FROM MOVIES2;
```

Solution:

	lang
►	Tamil
	Malayalam

4. Retrieve unique actor names.

```
SELECT DISTINCT act_name FROM MOVIES2;
```

Solution:

	act_name
▶	Kamal Haasan
	Mammootty
	Rajinikanth

Conclusion

MySQL is a powerful and widely used database management system. It provides an organized way to store and manage data using tables and SQL commands. Understanding database structure, DDL, DML, TCL, and different SQL data types provides a strong foundation for developing efficient and reliable database applications.